

Functional Annotation Report: scpC / PP_0154 (UniProt Q88RH5)

Pseudomonas putida KT2440 (PSEPK)

1. Summary (Answer to the Research Question)

scpC (PP_0154, Q88RH5) encodes a soluble, cytoplasmic class-I coenzyme-A transferase of the acetyl-CoA hydrolase/transferase family (Pfam PF02550 + PF13336; EC 2.8.3.-). It catalyzes the reversible transfer of coenzyme A between a short-chain acyl-CoA thioester and a free carboxylic acid (a transthioesterification), proceeding through a covalent glutamyl-CoA / acyl-glutamyl-anhydride intermediate on the active-site glutamate (Glu288). Its physiological substrate pair in *P. putida* has **not been measured directly**; the two authoritative computational annotations disagree: the EMBL/UniProt gene name "propionyl-CoA:succinate CoA transferase" is a homology transfer from *E. coli* YgfH/ScpC, whereas KEGG assigns it to orthology **K18118, succinyl-CoA:acetate CoA-transferase (EC 2.8.3.18, AarC-type)** in the TCA cycle. Both describe the same chemistry on cytoplasmic C2–C4 acyl-CoA/organic-acid pools feeding central carbon metabolism.

Identity verification (mandatory): The gene symbol *scpC* is ambiguous. This report concerns the *Pseudomonas putida* KT2440 **CoA-transferase** Q88RH5 (PP_0154). It is **not** the unrelated streptococcal *scpC* (a cell-envelope C5a peptidase/protease). The CoA-transferase identity is confirmed by the acetyl-CoA hydrolase/transferase domain architecture, the EC 2.8.3.- class, and orthology to *E. coli* ScpC/YgfH — all consistent with the UniProt record.

2. Molecular Identity and Structure

Property	Value
UniProt	Q88RH5
Gene / locus	<i>scpC</i> / PP_0154
Organism	<i>Pseudomonas putida</i> KT2440
Length / mass	497 aa / 53.96 kDa
Family	Acetyl-CoA hydrolase/transferase (class-I CoA-transferase)
Domains	PF02550 AcetylCoA_hydro (N, aa 11–215); PF13336 AcetylCoA_hyd_C (C, aa 322–463); SUPFAM SSF100950 (NagB/RpiA/CoA-transferase-like)
InterPro	IPR003702, IPR026888, IPR046433, IPR037171, IPR017821 (Succinate_CoA_transferase)
Orthology	eggNOG COG0427; KEGG K18118
Catalytic residue	Glu288 — "5-glutamyl coenzyme A thioester intermediate" (UniProt); binding residues 378/382/402
Localization	Cytoplasm (no signal peptide, no transmembrane segment)

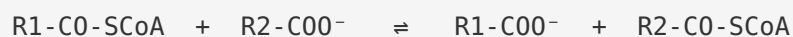
The two-domain $\alpha\beta$ architecture and the catalytic glutamate place *scpC* firmly in **class-I CoA-transferases**, which "catalyze transthioesterification reactions involving acyl-CoA substrates, using an active-site carboxylate to form covalent acyl anhydride and CoA thioester adducts" (Mullins & Kappock, 2012,).

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This mechanism is directly supported by UniProt's annotation of Glu288 as the glutamyl-CoA intermediate for Q88RH5.

3. Primary Function: The Catalyzed Reaction

scpC catalyzes a reversible, ping-pong CoA transfer:



Substrate specificity — two candidate assignments (unresolved for the *P. putida* enzyme):

1. **Propionyl-CoA:succinate CoA-transferase** (EMBL/UniProt name; EC 2.8.3.-). Based on the *E. coli* homolog: "YgfH is propionyl CoA:succinate CoA transferase" (Haller et al., 2000,).

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In *E. coli* this enzyme is the third step of the Sbm cycle converting succinate → propionate.

2. **Succinyl-CoA:acetate CoA-transferase** (KEGG K18118; EC 2.8.3.18). The best biochemically/structurally characterized member of this orthology group is *Acetobacter aceti* AarC: "AarC is succinyl-coenzyme A (CoA):acetate CoA-transferase, which replaces succinyl-CoA synthetase in a variant CAC" (Mullins et al., 2008,).

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Both activities are short-chain (C2–C4) acyl-CoA:carboxylate CoA-transfers by the same class-I mechanism; they differ only in the preferred acyl donor/acceptor pair. **No enzyme-kinetic study of the *P. putida* protein exists**, so its exact preference is inferred, not measured.

Sequence evidence (this work). Needleman–Wunsch pairwise alignment of Q88RH5 against characterized family members gives ~equal identity to the two succinate-linked transferases and low identity to a pure acetate transferase:

Homolog	Activity	% identity to scpC
<i>E. coli</i> YgfH/ScpC	propionyl-CoA:succinate CoA-transferase	48.3% (234/484)
<i>A. aceti</i> AarC	succinyl-CoA:acetate CoA-transferase (EC 2.8.3.18)	50.9% (251/493)
<i>E. coli</i> YdiF	acetate CoA-transferase	27.0% (129/477)

Because both nearest homologs share **succinate/succinyl-CoA** as one substrate, that partner is the most confidently assignable substrate for scpC. Identity is only marginally higher for AarC (+2.6%), so sequence cannot discriminate whether the second partner is propionyl-CoA (YgfH-like) or acetate (AarC-like); the slight AarC lean is consistent with the KEGG K18118 assignment.

Active-site integrity (this work). Residue mapping shows scpC's catalytic **Glu288** and CoA-binding residues **Asn378 / Gly382 / Lys402** are identical in both YgfH and AarC, confirming a complete, catalytically competent class-I active site (an active enzyme, not a pseudoenzyme).

These positions lie on the conserved CoA/catalytic side and therefore do not discriminate acceptor specificity. Homolog identities were independently verified from UniProt (P52043 = *E. coli* ScpC "Propanoyl-CoA:succinate CoA transferase"; A0AB33ICN5 = *A. aceti* AarC "Succinyl-CoA:acetate CoA-transferase"; P37766 = *E. coli* YdiF "Acetate CoA-transferase").

4. Pathway Context and Localization

- Localization:** cytoplasmic, soluble. The AarC-type CoA-transferase acts on the cytosolic organic-acid/acyl-CoA pool — the *A. aceti* *aar* genes "support flux through a complete but unorthodox CAC that is needed to lower cytoplasmic acetate levels" (P 18502856).
- Pathway (KEGG):** PP_0154 is mapped to the **citrate cycle (ppu00020)** and **pyruvate metabolism (ppu00620)**, consistent with a succinyl-CoA↔acetate transferase that can bypass succinyl-CoA synthetase.
- Genomic context distinguishes it from *E. coli*:** PP_0154 is a **standalone gene**. Its neighbors (PP_0153 hypothetical; PP_0152 exported protein; PP_0155/PP_0156 pyridine-nucleotide transhydrogenase) are unrelated to CoA/propionate metabolism. This contrasts with *E. coli*, where the homolog lies in a four-gene operon "that encodes homologues of methylmalonyl CoA mutases (Sbm) and acyl CoA transferases (YgfH)" (P 10769117).
- Plausible physiological niche (hypothesis).** No primary study characterizes PP_0154 in *P. putida*; its function is annotated by homology only. *P. putida* KT2440 assimilates acetate via acetyl-CoA and the glyoxylate shunt, and acetate-grown cells appear to use "ATP independent acetyl-CoA recycling as energy saving strategy" (Grether et al., 2026, P 42393666).

An AarC-type succinyl-CoA:acetate CoA-transferase (the KEGG K18118 activity) would provide exactly this ATP-independent activation of acetate to acetyl-CoA, consistent with PP_0154's KEGG mapping to the TCA cycle. This link is a reasonable but **untested** hypothesis.
- The Sbm route is absent in *P. putida*:** KEGG shows *P. putida* KT2440 encodes **no methylmalonyl-CoA mutase** (K01847) and no MCM subunit (K11942). Since propionate oxidation in *Pseudomonas* runs through the 2-methylcitrate cycle — "the only known and

described pathway for propionate oxidation in this organism" (Santos-Oliveira et al., 2024,

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— scpC cannot be part of an *E. coli*-style succinate→propionate cycle here. This argues that the "propionyl-CoA:succinate" name is a **homology over-transfer**, and that the KEGG succinyl-CoA:acetate CoA-transferase (TCA-linked) role is the more parsimonious annotation.

5. Supported and Refuted Hypotheses

Hypothesis	Verdict	Basis
scpC is a class-I CoA-transferase (acetyl-CoA hydrolase/transferase family) using a Glu covalent intermediate	Supported	Domain architecture, Glu288 active site, P 23030530
scpC catalyzes reversible short-chain acyl-CoA:carboxylate CoA transfer	Supported	EC 2.8.3.- class; homologs YgfH & AarC
scpC operates as part of an <i>E. coli</i> -style methylmalonyl-CoA (Sbm) succinate→propionate cycle in <i>P. putida</i>	Refuted	No methylmalonyl-CoA mutase/ decarboxylase in KT2440; gene not in an sbm operon
Exact substrate pair (propionyl-CoA/ succinate vs succinyl-CoA/acetate) is established for the <i>P. putida</i> enzyme	Unresolved	Conflicting database annotations; no direct kinetics
scpC is cytoplasmic	Supported (inferred)	No signal peptide/TM; analogy to soluble CAC enzymes

6. Limitations and Future Directions

- **No primary biochemistry exists for Q88RH5 itself.** Function is inferred from sequence/ family, orthology (KEGG K18118, COG0427), and characterized homologs (*E. coli* YgfH, *A. aceti* AarC). High-throughput/database annotations disagree on the exact acyl specificity.

- **Recommended experiments:** (i) purify recombinant PP_0154 and assay CoA-transfer with acetyl-/propionyl-/succinyl-CoA donors against acetate/succinate/propionate acceptors to define k^{cat}/K_m ; (ii) Glu288Ala mutant to confirm the catalytic nucleophile; (iii) $\Delta\text{PP_0154}$ phenotyping on acetate/propionate/succinate to place it physiologically; (iv) structure determination (or AlphaFold + docking) to map the auxiliary carboxylate-recognition site described for AarC.

References

- Haller T, Buckel T, Rétey J, Gerlt JA. *Biochemistry* 2000;39:4622-4629.

[P 10769117](#)

- Mullins EA, Francois JA, Kappock TJ. *J Bacteriol* 2008.

[P 18502856](#)

- Mullins EA, Kappock TJ. *Biochemistry* 2012.

[P 23030530](#)

- Froese DS et al. *Biochimie/...* 2009.

[P 18950999](#)

- Santos-Oliveira et al. 2024.

[P 37995793](#)

- Grether et al. *Acetate as alternative carbon source ... Pseudomonas putida KT2440*. 2026.

[P 42393666](#)

- Database records: UniProt Q88RH5; KEGG ppu:PP_0154 (K18118, EC 2.8.3.18); InterPro/Pfam domains as listed.
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Bottom Line

scpC/PP_0154 is a **soluble cytoplasmic, catalytically competent class-I CoA-transferase** (acetyl-CoA hydrolase/transferase family, EC 2.8.3.-) that performs reversible CoA transfer between a short-chain acyl-CoA thioester and a free carboxylic acid via a Glu288 glutamyl-CoA intermediate, feeding central carbon metabolism (TCA cycle). Succinate/succinyl-CoA is the most confidently assigned substrate partner; the identity of the second partner (propionyl-CoA vs acetate) is unresolved, with sequence, orthology (KEGG K18118), genomic context, and the absence of the Sbm pathway in *P. putida* collectively favouring a **succinyl-CoA:acetate CoA-transferase (AarC-type, EC 2.8.3.18)** role over the literal "propionyl-CoA:succinate" name. No gene-specific experimental data exist for the *P. putida* protein, so all functional statements are inferences from family, structure/sequence, and orthology.

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